

Skills Summary

Sign and Like-Term Errors in Polynomial Work

Use this reference while completing your worksheet or when studying.

Skill 1 — Distributing a minus sign

Rule: a minus sign in front of a bracket is a factor of -1 , so it changes the sign of *every* term inside:

$$A - (b + c - d) = A - b - c + d.$$

Rewrite the bracket with all its signs flipped *before* you collect anything. The term that gets missed is almost always the last one.

Example: Simplify $(6x + 5) - (2x - 8)$.

Step 1. A whole bracket is being subtracted, so my first move is to flip every sign inside it rather than start collecting.

Step 2. $-(2x - 8)$ becomes $-2x + 8$; the -8 turns into $+8$.

Step 3. Collect: $6x - 2x = 4x$ and $5 + 8 = 13$.

$$\Rightarrow 4x + 13$$

Try it: Simplify $(9x - 1) - (4x - 6)$.

check: $9x + x - 1 + 6$

Skill 2 — Combining only like terms

Rule: two terms are alike only when they have the *same variable to the same power*. Combining them adds the coefficients and leaves the power alone: $ax^n + bx^n = (a + b)x^n$. So $3x^2 + 4x$ is already as simple as it gets, and x^3 never joins x^2 .

Example: Simplify $4x^2 + 7x - x^2 + 2x$.

Step 1. Nothing is being subtracted as a bracket here, so the only question is which terms share a power.

Step 2. Squares: $4x^2 - x^2 = 3x^2$. Linear: $7x + 2x = 9x$.

$$\Rightarrow 3x^2 + 9x$$

Try it: Simplify $5x^3 + 2x - 3x^3 + 6x$.

check: $5x^3 - 3x^3 + 2x + 6x$

Skill 3 — Simplifying in stages, then checking

The test-value check: equivalent expressions agree at *every* number. So pick an easy value such as $x = 1$ or $x = 2$, substitute it into the original expression and into your simplified answer, and compare. One disagreement proves a mistake; agreement is strong evidence you are right. Avoid $x = 0$, which hides errors in the variable terms.

Example: Simplify $(2x^2 - x + 4) - (x^2 + 3x - 5)$, then check it at $x = 1$.

Step 1. A bracket is subtracted, so flip its signs first; then I will check the result with a test value instead of trusting the arithmetic.

Step 2. $2x^2 - x + 4 - x^2 - 3x + 5$, which collects to $x^2 - 4x + 9$.

Step 3. Test at $x = 1$: the original gives $(2 - 1 + 4) - (1 + 3 - 5) = 5 - (-1) = 6$, and the answer gives $1 - 4 + 9 = 6$.

$\Rightarrow$ $x^2 - 4x + 9$, and both sides give **6** at $x = 1$.

Try it: Simplify $(5x^2 + 2x - 3) - (2x^2 - 4x + 1)$.

check: $1 - 4 + 9 = 6$